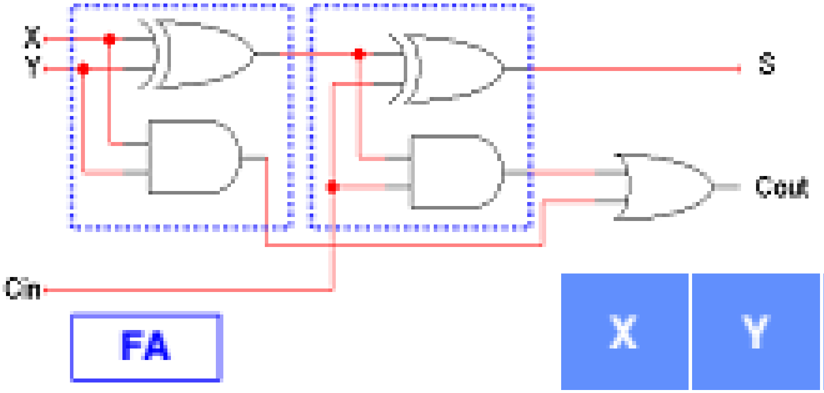
Carry Look Ahead Adder

- Instead of waiting for the carry from previous stages, a carry look ahead adder computes carry directly from input with a two level circuit, minimizing delay
- In Carry Look Ahead Adder circuit , carry's entering into all bit positions are computed simultaneously by a carry look ahead generator
- Carry computation takes place in parallel with sum computation
- Carry look ahead generator results in a constant addition time irrespective of the length of the adder

Full Adder



For a Full Adder

S = X ⊕ Y ⊕ Cin
Cout = XY + Cin(X ⊕ Y)

X	Y	Cin	Sum	Cout	Carry Status
0	0	0	0	0	Delete
0	0	1	1	0	Delete
0	1	0	1	0	Propagate
0	1	1	0	1	Propagate
1	0	0	1	0	Propagate
1	0	1	0	1	Propagate
1	1	0	0	1	Generate
1	1	1	1	1	Generate

Carry Propagate & Generate

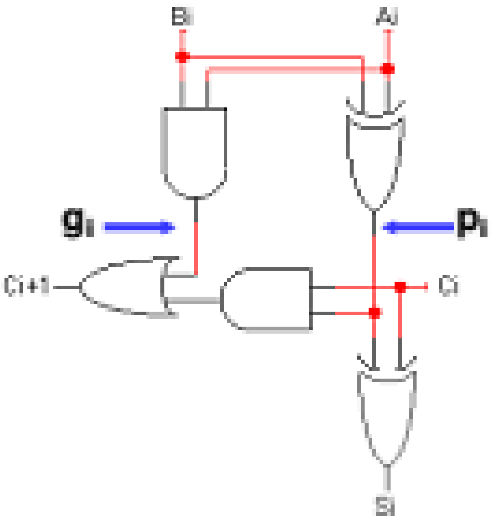
- **Carry Generate**
  - When both inputs A & B are 1, Cout is generated i.e. 1, irrespective of Cin. Carry generation is denoted by 'g'

$$g_i = A_i B_i$$

- **Carry Propagate**
  - When one of the inputs A & B is 1 but not both, carry is propagated i.e. Cout = Cin. Carry propagation is denoted by 'p'

$$p_i = A_i \oplus B_i$$

- Carry Out Expression
  - $c_{i+1} = g_i + p_i c_i$



'p' & 'g' computation has one level of gate delay

Algebraic Carry out Expressions

- Carry out equations for specific bits, using the general equation $c_{i+1} = g_i + p_i c_i$

$c_1 = g_0 + p_0 c_0$

$c_2 = g_1 + p_1 c_1$
$= g_1 + p_1(g_0 + p_0 c_0)$
$= g_1 + p_1 g_0 + p_1 p_0 c_0$

$c_3 = g_2 + p_2 c_2$
$= g_2 + p_2(g_1 + p_1 g_0 + p_1 p_0 c_0)$
$= g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_0$

$c_4 = g_3 + p_3 c_3$
$= g_3 + p_3(g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_0)$
$= g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0 + p_3 p_2 p_1 p_0 c_0$

Cin = c_0

All Carry Expressions are in Sum of product form

All Carry Expressions are only dependent on Cin in addition to ‘p’ & ‘g’ functions

- Carry expressions logic has two level gate delay after ‘p’ & ‘g’ are computed

Algebraic Sum Expressions

- Sum expression for a full adder circuit is
  - $S_0 = a_0 \oplus b_0 \oplus c_0$
  - $S_0 = p_0 \oplus c_0$
- Sum Expression for Specific bits are

$$s_0 = p_0 \oplus c_0$$

$$s_1 = p_1 \oplus c_1$$

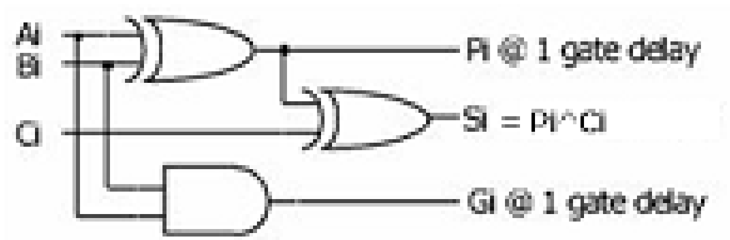
$$s_2 = p_2 \oplus c_2$$

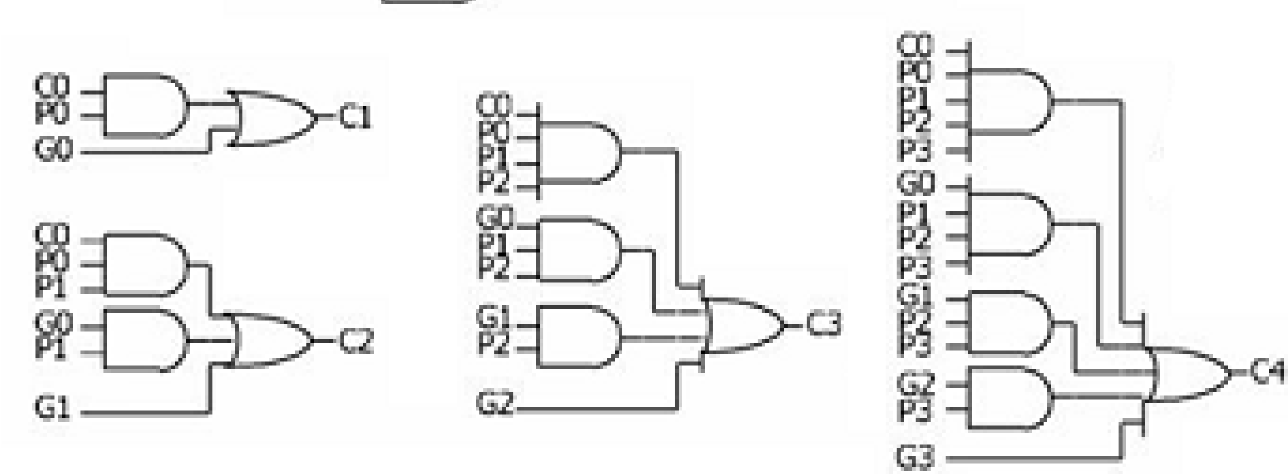
$$s_3 = p_3 \oplus c_3$$

- After Carries are computed , Sum computation has one level of gate delay

Carry Circuits Implementation

- **Carry circuit implementation in terms of carry propagate and generate**





- **Carry circuit implementation gets complicated with increase in bit positions**

4-bit Carry look-ahead Adder